

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1829

Upper Lake 138 kV
August 23, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1829. The Interconnection Customer proposes a 149.3 MW solar plus storage facility interconnecting at the Upper Lake 138 kV substation in the Virginia area. The request was received on July 25, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 422 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$3,260,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Pound – Upper Lake 138 KV L	Toano – Surry 345 KV	100.7	212	68.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Pound – Upper Lake 138 KV L	Reconductor line	\$5,390,000	60% (\$3,260,000)
	Total allocated to project		\$3,260,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.